# Couplings, Wasserstein Distances, and Kullback-Leibler Divergence

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**Abstract**

We study two prominent ways of measuring discrepancy between probability distributions—the Kullback-Leibler (KL) divergence and the Wasserstein distance—on a shared compact metric space $X$. After defining each object precisely and cataloguing its key properties, we examine how they differ in geometric sensitivity, in their requirements on absolute continuity, and in their dual variational representations. A central question is whether the KL divergence, like the Wasserstein distance, can be understood as an optimisation over *couplings* (joint distributions with prescribed marginals). We argue that it cannot in general—KL uses a fixed pointwise comparison at the same location $x \in X$ and involves no transport of mass—but that the two objects are linked through two related constructions: *entropy-regularised optimal transport* (in which KL appears as a penalty on couplings) and the classical *Schrödinger bridge* problem (in which KL on joint distributions replaces the transport cost). We close with inequalities that bound one quantity in terms of the other, including Pinsker's inequality and Talagrand's transport inequality.

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# 1 Introduction

Let $X$ be a compact metric space equipped with its Borel $\sigma$-algebra $\mathcal{B}(X)$ and a reference measure $\lambda$ (typically Lebesgue measure on a subset of $\mathbb{R}^d$, or the counting measure on a finite set). Denote by $\mathcal{P}(X)$ the set of Borel probability measures on $X$.

We consider two probability density functions $f$ and $g$ with respect to $\lambda$, so that $f, g \geq 0$ and $\int_X f\, d\lambda = \int_X g\, d\lambda = 1$. We say that $f$ and $g$ are *mutually absolutely continuous* if $f(x) = 0 \Leftrightarrow g(x) = 0$ $\lambda$-almost everywhere, written $f \ll g$ and $g \ll f$. In the language of measures, letting $\mu, \nu \in \mathcal{P}(X)$ denote the measures with densities $f$ and $g$ respectively, mutual absolute continuity means $\mu \ll \nu$ and $\nu \ll \mu$.

Two fundamental and ubiquitous ways to measure how far $f$ is from $g$ are:

1. The **Kullback-Leibler divergence** $D_{\mathrm{KL}}(f \,\|\, g)$, rooted in information theory and statistics.
2. The **Wasserstein distance** $W_p(f,\, g)$, rooted in optimal transport theory.

Both reduce to zero if and only if $f = g$ almost everywhere. Beyond this they are, in many ways, measuring different things: KL divergence is insensitive to the ground metric $d$ on $X$ (the metric on the *sample space*); Wasserstein is exquisitely sensitive to it. On the other hand, KL divergence does induce a rich geometry on the *space of probability distributions* $\mathcal{P}(X)$ via the Fisher information metric—the subject of Amari's information geometry [Amari, 2016, Amari and Nagaoka, 2000]—a distinct geometry from that generated by $W_2$. KL divergence requires absolute continuity; Wasserstein does not. KL divergence is defined directly in terms of the two densities evaluated at the same point; Wasserstein is defined as an infimum over all joint distributions (*couplings*) of a pair having $f$ and $g$ as their marginals.

Understanding these similarities and differences is essential for applications in statistics, machine learning, information theory, mathematical physics, and economics. This note aims to give a self-contained exposition.

# 2 Kullback-Leibler Divergence

## 2.1 Definition

$$\boxed{D_{\mathrm{KL}}(f \,\|\, g) = \int_X f(x) \log \frac{f(x)}{g(x)} \, d\lambda(x).} \tag{1}$$

Here the integrand is defined to be 0 when $f(x) = 0$ (by $0 \log 0 = 0$), and $+\infty$ when $f(x) > 0 = g(x)$. Thus $D_{\mathrm{KL}}(f \,\|\, g) \in [0, +\infty]$, and it is finite precisely when $f \ll g$. Under mutual absolute continuity, it is always finite.

The KL divergence can be decomposed as

$$D_{\mathrm{KL}}(f \,\|\, g) = -H(f) + H_{\mathrm{cross}}(f, g), \tag{2}$$

where $H(f) = -\int f \log f \, d\lambda$ is the *differential entropy* of $f$, and $H_{\mathrm{cross}}(f, g) = -\int f \log g \, d\lambda$ is the *cross-entropy* of $f$ relative to $g$.

## 2.2 Properties

**Non-negativity (Gibbs' inequality).** $D_{\mathrm{KL}}(f \,\|\, g) \geq 0$, with equality if and only if $f = g$ $\lambda$-a.e. This follows from Jensen's inequality applied to the strictly convex function $\phi(t) = t \log t$:

$$D_{\mathrm{KL}}(f \,\|\, g) = \int_X g(x) \, \phi\left(\frac{f(x)}{g(x)}\right) d\lambda(x) \geq \phi\left(\int_X g(x) \frac{f(x)}{g(x)} \, d\lambda(x)\right) = \phi(1) = 0. \tag{3}$$

**Asymmetry.** In general $D_{\mathrm{KL}}(f \,\|\, g) \neq D_{\mathrm{KL}}(g \,\|\, f)$. The *symmetrised KL divergence* (also called the $J$-divergence) is

$$J(f, g) = D_{\mathrm{KL}}(f \,\|\, g) + D_{\mathrm{KL}}(g \,\|\, f) = \int_X \big(f(x) - g(x)\big) \log \frac{f(x)}{g(x)} \, d\lambda(x), \tag{4}$$

which is symmetric and non-negative.

**Not a metric.** $D_{\mathrm{KL}}(f \,\|\, g)$ is not a metric: it is not symmetric, and it does not satisfy the triangle inequality.

**Convexity.** The map $(f, g) \mapsto D_{\mathrm{KL}}(f \,\|\, g)$ is jointly convex: for any $\alpha \in [0, 1]$,

$$D_{\mathrm{KL}}(\alpha f_1 + (1-\alpha) f_2 \,\|\, \alpha g_1 + (1-\alpha) g_2) \leq \alpha\, D_{\mathrm{KL}}(f_1 \,\|\, g_1) + (1-\alpha)\, D_{\mathrm{KL}}(f_2 \,\|\, g_2)\,.$$

**Data processing inequality.** Let $T : X \to Y$ be any measurable map and let $T_{\#}\mu$, $T_{\#}\nu$ denote the push-forward measures. Then

$$D_{\mathrm{KL}}(T_{\#}\mu \,\|\, T_{\#}\nu) \leq D_{\mathrm{KL}}(\mu \,\|\, \nu)\,. \tag{5}$$

Equivalently, processing the data reduces (or preserves) information.

**Chain rule.** Let $\pi_f$ and $\pi_g$ be joint distributions over $X \times X$ with $x$-marginals $f$ and $g$ respectively. Denote by $f(\cdot|x)$ and $g(\cdot|x)$ their conditional distributions given the first coordinate. Then

$$D_{\mathrm{KL}}(\pi_f \,\|\, \pi_g) = D_{\mathrm{KL}}(f_1 \,\|\, g_1) + \int_X f_1(x)\, D_{\mathrm{KL}}(f(\cdot|x) \,\|\, g(\cdot|x))\; d\lambda(x), \tag{6}$$

where $f_1$ and $g_1$ are the first marginal densities. The chain rule shows that KL divergence is additive over conditional levels of a hierarchy.

**Sensitivity to tails.** Because the integrand in (1) is $f \log(f/g) = f \log f - f \log g$, regions where $g$ is very small but $f$ is not contribute large (potentially infinite) quantities. KL divergence is thus very sensitive to tail behaviour. The Wasserstein distance, by contrast, is insensitive to extreme tails (on a compact space, it is always finite).

## 2.3 Variational Representation

The *Donsker-Varadhan variational formula* [Donsker and Varadhan, 1975] expresses KL divergence as the Legendre transform of the log-moment-generating functional:

$$\boxed{D_{\mathrm{KL}}(f \,\|\, g) = \sup_{h \in L^1(f)} \left[\mathbb{E}_f[h(X)] - \log \mathbb{E}_g\left[e^{h(X)}\right]\right].} \tag{7}$$

The supremum is attained at the function $h^*(x) = \log\big(f(x)/g(x)\big)$. Substituting $h^*$:

$$\mathbb{E}_f[h^*] - \log \mathbb{E}_g\left[e^{h^*}\right] = \mathbb{E}_f\left[\log \frac{f}{g}\right] - \log \int g \cdot \frac{f}{g}\, d\lambda = D_{\mathrm{KL}}(f \,\|\, g) - \log 1 = D_{\mathrm{KL}}(f \,\|\, g)\,. \checkmark$$

The formula (7) is extremely useful because the right-hand side depends only on expectations under $f$ and $g$ separately, with no need to evaluate the ratio $f/g$ directly. This makes it numerically tractable and is the basis of many estimators of KL divergence from samples.

## 2.4 The $f$-Divergence Family

KL divergence is a member of the *$f$-divergence* family introduced by Csiszár [1975]. Given a convex function $\phi : (0, \infty) \to \mathbb{R}$ with $\phi(1) = 0$, the $\phi$-divergence is

$$D_\phi(f\|g) = \int_X g(x)\, \phi\left(\frac{f(x)}{g(x)}\right) d\lambda(x). \tag{8}$$

Special cases are listed in Table 1.

Table 1: Members of the $f$-divergence family defined by (8). All satisfy $D_\phi \geq 0$ with equality iff $f = g$ a.e.

| Divergence | Generator $\phi(t)$ | Formula |
|---|---|---|
| KL (forward) | $t \log t$ | $\int f \log(f/g)$ |
| KL (reverse) | $-\log t$ | $\int g \log(g/f)$ |
| Total variation | $\frac{1}{2}\lvert t-1\rvert$ | $\frac{1}{2}\int \lvert f-g\rvert$ |
| $\chi^2$-divergence | $(t-1)^2$ | $\int (f-g)^2/g$ |
| Hellinger squared | $(\sqrt{t}-1)^2$ | $1-\int \sqrt{fg}$ |
| Jensen–Shannon | $(t+1)\log\frac{2}{t+1}+t\log t$ | $\frac{1}{2}[D_{\mathrm{KL}}(f \,\|\, m) + D_{\mathrm{KL}}(g \,\|\, m)]$, $m = \frac{f+g}{2}$ |

$f$-divergences share several structural features: non-negativity, joint convexity, and the data-processing inequality (5). Crucially for our comparison, they are all *reference-measure-based*: they evaluate the ratio $f(x)/g(x)$ at the same point $x$. No transport is involved.

# 3 Couplings

## 3.1 Marginals and the Set of Couplings

Let $\mu$ and $\nu$ be two probability measures in $\mathcal{P}(X)$. A *coupling* of $\mu$ and $\nu$ is a probability measure $\pi \in \mathcal{P}(X \times X)$ whose first marginal is $\mu$ and whose second marginal is $\nu$:

$$\pi(A \times X) = \mu(A) \quad \text{and} \quad \pi(X \times B) = \nu(B) \quad \text{for all } A, B \in \mathcal{B}(X). \tag{9}$$

The set of all couplings is denoted

$$\Pi(\mu, \nu) = \big\{\pi \in \mathcal{P}(X \times X) : (9) \text{ holds}\big\}. \tag{10}$$

$\Pi(\mu, \nu)$ is always non-empty (it contains the *independent coupling* $\mu \otimes \nu$ defined by $(\mu \otimes \nu)(A \times B) = \mu(A)\nu(B)$), convex, and compact in the weak topology when $X$ is compact.

If $\mu$ and $\nu$ both have densities $f$ and $g$ with respect to $\lambda$, a coupling $\pi \in \Pi(\mu, \nu)$ with joint density $\pi(x, y)$ satisfies:

$$\int_X \pi(x, y)\, d\lambda(y) = f(x), \qquad \int_X \pi(x, y)\, d\lambda(x) = g(y). \tag{11}$$

These are the *marginal constraints* of the optimal transport problem.

## 3.2 Examples of Couplings

**1. The independent coupling.** $\pi_{\mathbf{1}}(x, y) = f(x)g(y)$. Under this coupling, $X$ and $Y$ are independent: knowing $X = x$ conveys no information about $Y$. The independent coupling is the unique maximiser of the entropy $H(\pi) = -\int \pi \log \pi$ subject to the marginal constraints.

**2. Deterministic (Monge) couplings.** Given a *transport map* $T : X \to X$ with $T_{\#}\mu = \nu$ (i.e., $\nu(B) = \mu(T^{-1}(B))$ for all $B$), the corresponding coupling is

$$\pi_T(A \times B) = \mu(A \cap T^{-1}(B)). \tag{12}$$

In terms of density: $\pi_T(x, y) = f(x)\,\delta(y - T(x))$. Under this coupling, $Y = T(X)$ is a deterministic function of $X$. The marginal constraint on $y$ requires $T_{\#}\mu = \nu$, which is the condition defining a valid transport map.

**3. The diagonal coupling.** When $\mu = \nu$ (i.e., $f = g$), the *diagonal coupling* $\pi_\Delta(x, y) = f(x)\delta(y - x)$ has both marginals equal to $f$. Under this coupling $X = Y$ almost surely. Note that the diagonal coupling is a special case of a Monge coupling with $T = \mathrm{id}$, and it only belongs to $\Pi(\mu, \nu)$ when $\mu = \nu$.

**4. An anti-monotone coupling.** On $X = [0, 1]$, pair the $\alpha$-quantile of $f$ with the $(1 - \alpha)$-quantile of $g$ for each $\alpha \in [0, 1]$. This coupling is "anti-aligned" and in general incurs larger transport cost than the identity coupling; it is used as a worst-case comparison.

**5. The optimal coupling.** For a given cost function $c : X \times X \to [0, \infty)$, the coupling that minimises $\int c\, d\pi$ over $\Pi(\mu, \nu)$ is the *optimal (Kantorovich) coupling*. Its existence and characterisation is the central object of optimal transport theory, to which we now turn.

# 4 Optimal Transport

## 4.1 The Monge Problem

Gaspard Monge posed the original mass-transportation problem in 1781: find a measurable map $T : X \to X$ that transports $\mu$ to $\nu$ while minimising the total cost:

$$\inf_{T:\, T_{\#}\mu = \nu} \int_X c(x, T(x))\, d\mu(x). \tag{13}$$

Here $c : X \times X \to [0, \infty)$ is a *cost function* measuring the expense of moving a unit of mass from $x$ to $T(x)$. Typical choices are $c(x, y) = d(x, y)$ (transport distance proportional to displacement) or $c(x, y) = d(x, y)^2$ (quadratic cost, penalising large displacements more severely).

The Monge problem has two difficulties. First, the feasible set $\{T : T_{\#}\mu = \nu\}$ may be empty: for example, if $\mu$ is a Dirac mass $\delta_{x_0}$ and $\nu$ is not, there is no map sending $\mu$ to $\nu$. Second, even when a map exists, the infimum may not be attained.

## 4.2 The Kantorovich Relaxation

Kantorovich [1942] relaxed the Monge problem by allowing *mass splitting*: instead of a deterministic map, seek a coupling $\pi \in \Pi(\mu, \nu)$ that minimises the expected cost:

$$\boxed{C(\mu, \nu; c) = \inf_{\pi \in \Pi(\mu,\nu)} \int_{X \times X} c(x, y)\, d\pi(x, y).} \tag{14}$$

The Kantorovich problem is a *linear programme* over the convex set $\Pi(\mu, \nu)$, and it always has a solution when $c$ is lower semicontinuous and $X$ is compact. Every Monge coupling is a special case (it restricts $\pi$ to the set $\{(x, T(x)) : x \in X\}$), so the Kantorovich infimum is at most the Monge infimum. On well-behaved spaces the two coincide.

## 4.3 Kantorovich Duality

The Kantorovich problem has a beautiful dual formulation. Define a pair of functions $(\phi, \psi) : X \times X \to \mathbb{R}$ to be *c-admissible* if $\phi(x) + \psi(y) \leq c(x, y)$ for all $(x, y)$. The *Kantorovich dual* is:

$$C(\mu, \nu; c) = \sup_{\phi+\psi \leq c} \left[ \int_X \phi(x)\, d\mu(x) + \int_X \psi(y)\, d\nu(y) \right]. \tag{15}$$

Equality between (14) and (15) (strong duality) holds under mild conditions [Villani, 2009]. The dual variables $\phi$ and $\psi$ are called *Kantorovich potentials*; they play the role of prices in an equilibrium interpretation.

For the quadratic cost $c(x, y) = |x - y|^2$ on $\mathbb{R}^d$, the $c$-transform $\phi^c(y) = \inf_x[|x - y|^2 - \phi(x)]$ yields $\psi = \phi^c$ at the optimum, and both $\phi$ and $-\phi^c$ are convex functions. Brenier's theorem [Brenier, 1991] then gives: if $\mu$ is absolutely continuous, the optimal map is $T = \nabla\phi$ for a convex function $\phi$—a gradient of convex potential.

# 5 Wasserstein Distance

## 5.1 Definition

Let $(X, d)$ be a compact metric space and fix $p \geq 1$. The *Wasserstein-p distance* between $\mu$ and $\nu$ in $\mathcal{P}(X)$ is

$$\boxed{W_p(\mu, \nu) = \left( \inf_{\pi \in \Pi(\mu,\nu)} \int_{X \times X} d(x, y)^p \, d\pi(x, y) \right)^{1/p}.} \tag{16}$$

This is the Kantorovich cost (14) with $c(x, y) = d(x, y)^p$, raised to the power $1/p$. Because $X$ is compact, the infimum is attained and $W_p(\mu, \nu) < \infty$ for all $\mu, \nu \in \mathcal{P}(X)$.

## 5.2 Metric Properties

$W_p$ is a metric on $\mathcal{P}(X)$:

- **Non-negativity:** $W_p \geq 0$, with $W_p(\mu, \nu) = 0$ iff $\mu = \nu$.
- **Symmetry:** $W_p(\mu, \nu) = W_p(\nu, \mu)$, because coupling is symmetric.
- **Triangle inequality:** $W_p(\mu, \nu) \leq W_p(\mu, \sigma) + W_p(\sigma, \nu)$ for all $\sigma$. This follows from the *gluing lemma*: given optimal couplings $\pi_1 \in \Pi(\mu, \sigma)$ and $\pi_2 \in \Pi(\sigma, \nu)$, one can construct a joint distribution on $X^3$ whose projections on the relevant pairs yield a coupling in $\Pi(\mu, \nu)$.

The metric $W_p$ metrises weak convergence of probability measures on $X$ (when $X$ is compact, this is equivalent to convergence in distribution).

## 5.3 Ordering Across $p$

By Jensen's inequality applied to the concave function $t \mapsto t^{p/q}$ for $p \leq q$:

$$W_p(\mu, \nu) \leq W_q(\mu, \nu), \quad 1 \leq p \leq q. \tag{17}$$

Thus $W_1 \leq W_2 \leq W_\infty$ and smaller $p$ gives a weaker topology.

## 5.4 The Case $p = 1$: Kantorovich-Rubinstein

For $p = 1$ the Kantorovich dual (15) with $c = d$ simplifies to the celebrated *Kantorovich-Rubinstein* representation:

$$\boxed{W_1(\mu, \nu) = \sup_{\|h\|_{\mathrm{Lip}} \le 1} \left[ \int_X h \, d\mu - \int_X h \, d\nu \right],} \tag{18}$$

where the supremum is over all 1-Lipschitz functions $h : X \to \mathbb{R}$ (i.e., $|h(x) - h(y)| \le d(x, y)$). $W_1$ is also called the *earth mover's distance* because it measures the minimum work needed to redistribute a pile of earth from shape $\mu$ to shape $\nu$.

## 5.5 The Case $p = 2$: Brenier Maps and Displacement Interpolation

For $p = 2$ on $\mathbb{R}^d$ (or on a Riemannian manifold), the optimal coupling under the quadratic cost is, under mild regularity, a deterministic coupling induced by the *Brenier map* $T = \nabla\phi$ (gradient of a convex potential $\phi$): the unique optimal map $T$ such that $T_{\#}\mu = \nu$ [Brenier, 1991].

A remarkable feature of $W_2$ is that it supports a Riemannian-like geometry on $\mathcal{P}(X)$. The *displacement interpolation* [McCann, 1997] $\mu_t = ((1 - t)\,\mathrm{id} + t\,T)_{\#}\mu$ is the geodesic (shortest path) between $\mu$ and $\nu$ in the $W_2$ metric. This provides a natural notion of "straight-line interpolation" between probability distributions, with no analogue in the KL framework.

The $W_2$ metric also arises naturally in the *Jordan-Kinderlehrer-Otto (JKO) scheme* [Jordan et al., 1998]: the Fokker-Planck equation for a diffusion process is the gradient flow of the free energy functional $\mu \mapsto D_{\mathrm{KL}}(\mu \,\|\, \mu_\infty)$ (where $\mu_\infty$ is the invariant measure) with respect to the $W_2$ metric. This deep connection between gradient flows, KL divergence, and $W_2$ is the subject of Ambrosio et al. [2008].

## 5.6 Summary of Properties

Table 2 collects key properties for comparison.

Table 2: Properties of $D_{\mathrm{KL}}(f \,\|\, g)$ and $W_p(\mu, \nu)$, where $f, g$ are mutually absolutely continuous densities on a compact metric space $(X, d)$.

| Property | $D_{\mathrm{KL}}(f \,\|\, g)$ | $W_p(\mu, \nu)$ |
|---|---|---|
| Non-negative | Yes | Yes |
| Symmetric | No | Yes |
| Triangle inequality | No | Yes |
| Metric | No (divergence only) | Yes |
| Requires $f \ll g$ for finiteness | Yes | No |
| Sensitive to ground metric $d$ on $X$ | No | Yes |
| Induces geometry on $\mathcal{P}(X)$ | Yes (Fisher–Rao) | Yes (Otto) |
| Defined via coupling | No | Yes |
| Dual representation | Donsker–Varadhan | Kantorovich–Rubinstein |
| Convex in $(f, g)$ jointly | Yes | Yes (convex in each marginal) |
| Gradient-flow structure | KL = functional in $W_2$-GF | $W_2$ = metric |

# 6 Comparing Kullback-Leibler and Wasserstein

## 6.1 Sensitivity to the Ground Metric on $X$

A key difference between KL divergence and Wasserstein distance is their relationship to the *ground metric* $d$ on the sample space $X$—to be distinguished carefully from the geometry each object induces on the *space of probability distributions* $\mathcal{P}(X)$ (which we discuss in Section 6.2).

The Wasserstein distance $W_p(\mu, \nu)$ is constructed from the ground metric $d$ on $X$: moving mass from $x$ to $y$ costs $d(x, y)^p$. Consequently, if $\mu = \delta_{x_0}$ and $\nu = \delta_{y_0}$ are point masses, then $W_p(\mu, \nu) = d(x_0, y_0)^p > 0$ whenever $x_0 \neq y_0$, and $W_p \to 0$ as $d(x_0, y_0) \to 0$. The Wasserstein distance encodes the ground metric.

The KL divergence does not use $d$ at all. In the same example, $\mu = \delta_{x_0}$ and $\nu = \delta_{y_0}$ are not absolutely continuous with respect to each other ($\delta_{x_0} \not\ll \delta_{y_0}$), so $D_{\mathrm{KL}}(\mu \,\|\, \nu) = +\infty$ regardless of how small $d(x_0, y_0)$ is.

As a continuous example, consider on $X = [0, 1]$ the densities $f_n(x) = 1 + \sin(2\pi n x)$ and $g_n(x) = 1 - \sin(2\pi n x)$. Both are valid probability densities for all $n \geq 1$. As $n \to \infty$,

the two oscillate faster and faster; they become equally spread over $[0, 1]$ but always "out of phase." It can be shown that $W_1(f_n, g_n) \to 0$ (the mass only needs to travel $\approx 1/n$ to cancel the oscillations), while $D_{\mathrm{KL}}(f_n \,\|\, g_n)$ remains bounded away from zero (the pointwise ratio $f_n/g_n$ continues to oscillate significantly).

## 6.2 Information Geometry: The Geometry KL Induces on $\mathcal{P}(X)$

Although KL divergence ignores the ground metric $d$ on $X$, it is far from geometry-free: it *generates* a Riemannian geometry on the statistical manifold $\mathcal{P}(X)$ through its infinitesimal expansion. This is the subject of *information geometry* [Amari, 1985, Amari and Nagaoka, 2000, Amari, 2016, Nielsen, 2020].

Let $\mathcal{M} = \{p(\cdot;\theta) : \theta \in \Theta\}$ be a smooth parametric family of probability densities. A straightforward Taylor expansion of the KL divergence in a small perturbation $\theta \to \theta + d\theta$ gives:

$$D_{\mathrm{KL}}(p(\cdot;\theta) \,\|\, p(\cdot;\theta + d\theta)) = \tfrac{1}{2}\, g_{ij}(\theta)\, d\theta^i d\theta^j + O(|d\theta|^3), \tag{19}$$

where the coefficients

$$g_{ij}(\theta) = \mathbb{E}_\theta\left[\frac{\partial \log p(X;\theta)}{\partial \theta_i}\frac{\partial \log p(X;\theta)}{\partial \theta_j}\right] = -\mathbb{E}_\theta\left[\frac{\partial^2 \log p(X;\theta)}{\partial \theta_i \partial \theta_j}\right] \tag{20}$$

form the *Fisher information matrix*. The matrix $(g_{ij}(\theta))$ is positive semidefinite (positive definite under identifiability conditions) and defines a Riemannian metric on $\mathcal{M}$—the *Fisher-Rao metric* [Rao, 1945]. In this sense, the KL divergence is the *potential* whose Hessian generates the geometry of the statistical manifold.

Amari [Amari, 1985, Amari and Nagaoka, 2000] enriched this Riemannian structure with a family of *$\alpha$-connections* $\nabla^{(\alpha)}$ for $\alpha \in [-1, 1]$. The $e$-connection ($\alpha = +1$, exponential) and the $m$-connection ($\alpha = -1$, mixture) are dually flat with respect to the Fisher metric. These connections encode the natural coordinate systems for exponential and mixture families respectively—and the KL divergence is the *divergence function* that generates the dual structure via:

$$D_{\mathrm{KL}}(p_0 \,\|\, p_1) = \tfrac{1}{2}\, g_{ij}\, d\theta^i d\theta^j - \tfrac{1}{3} T_{ijk}\, d\theta^i d\theta^j d\theta^k + \cdots,$$

where $T_{ijk}$ (the *Amari-Chentsov tensor*) encodes the skewness of the deviation from Riemannian symmetry.

A central operation in information geometry is the *m-projection* (mixture projection) and *e-projection* (exponential projection):

$$\Pi_e(p;\mathcal{F}) = \arg\min_{q\in\mathcal{F}} D_{\mathrm{KL}}(p \,\|\, q)\,, \qquad \Pi_m(p;\mathcal{F}) = \arg\min_{q\in\mathcal{F}} D_{\mathrm{KL}}(q \,\|\, p)\,,$$

for a sub-manifold $\mathcal{F}$. The Pythagorean theorem of information geometry states that for dually flat spaces and a convex sub-manifold $\mathcal{F}$:

$$D_{\mathrm{KL}}(p \,\|\, q) = D_{\mathrm{KL}}(p \,\|\, \Pi_e(p;\mathcal{F})) + D_{\mathrm{KL}}(\Pi_e(p;\mathcal{F}) \,\|\, q)\,, \qquad \forall\, q \in \mathcal{F},$$

an exact analogue of the Euclidean Pythagorean theorem. Frank Nielsen and collaborators [Nielsen, 2020, Nielsen and Nock, 2010] have developed efficient computational algorithms exploiting this geometry, including Bregman-divergence-based Voronoi diagrams and clustering in the KL geometry.

**Two distinct geometries on $\mathcal{P}(X)$.** The Fisher-Rao geometry and the Wasserstein-2 (Otto) geometry are *both* valid Riemannian geometries on the space of probability distributions, but they capture different phenomena:

- The **Fisher-Rao geometry** is intrinsic to the *distributional family*: it measures infinitesimal distinguishability between neighbouring distributions, independently of any metric structure on the sample space $X$. Geodesics in the Fisher-Rao metric are statistically natural (e.g., for exponential families they are straight lines in natural parameters).
- The **Otto (Wasserstein-2) geometry** is built from the ground metric $d$ on $X$: it measures the cost of transporting mass between distributions. Geodesics in the $W_2$ metric are displacement interpolations (Section 5).

The two geometries are genuinely different: the Fisher-Rao geodesic between two Gaussians passes through other Gaussians, while the $W_2$ geodesic (displacement interpolation) generates intermediate distributions that are also Gaussian but follow a different path through parameter space [Villani, 2009].

## 6.3 Absolute Continuity

$D_{\mathrm{KL}}(f \,\|\, g)$ requires $\mu \ll \nu$ (i.e., $f \ll g$) to be finite. If $\mu$ assigns positive mass to a set that $\nu$ assigns zero mass, then $D_{\mathrm{KL}}(\mu \,\|\, \nu) = +\infty$.

$W_p(\mu, \nu)$ is defined and finite for all $\mu, \nu \in \mathcal{P}(X)$ when $X$ is compact, with no absolute-continuity requirement. The optimal coupling $\pi^*$ may or may not have a density.

This distinction has important practical consequences. When learning a generative model, if the model distribution $\mu_\theta$ has support on a lower-dimensional manifold (e.g., a neural network maps Gaussian noise to images), then $D_{\mathrm{KL}}(\mu_\theta \,\|\, \nu)$ may be $+\infty$ for most $\theta$, giving no useful gradient. The Wasserstein distance is well-defined in these settings and provides useful gradient information—the key observation behind the Wasserstein GAN [Peyré and Cuturi, 2019, Ch. 7].

## 6.4 Dual Representations Compared

The dual representations of $W_1$ and $D_{\mathrm{KL}}(f \,\|\, g)$ are worth placing side by side:

$$W_1(\mu, \nu) = \sup_{\|h\|_{\mathrm{Lip}} \le 1} \Big[ \mathbb{E}_\mu[h] - \mathbb{E}_\nu[h] \Big], \tag{21}$$

$$D_{\mathrm{KL}}(f \,\|\, g) = \sup_{h \in L^1(\mu)} \Big[ \mathbb{E}_\mu[h] - \log \mathbb{E}_\nu \big[ e^h \big] \Big]. \tag{22}$$

Both involve the difference of expectations of a test function $h$ under the two measures. The differences are:

- $W_1$ uses a *linear* test-function difference $\mathbb{E}_\mu[h] - \mathbb{E}_\nu[h]$, constrained to Lipschitz functions. The constraint encodes the geometry of $X$ through the Lipschitz condition.
- KL uses a *non-linear* expression $\mathbb{E}_\mu[h] - \log \mathbb{E}_\nu \big[ e^h \big]$. The exponential transforms $h$ by an exponential twist of the measure $\nu$ (a Cramér tilt), which is sensitive to large deviations in the tails. There is no geometric constraint on $h$.

The linear structure of $W_1$'s dual makes it well-behaved for stochastic gradient methods and stable under small geometry- preserving perturbations. The exponential structure of KL's dual makes it powerful for hypothesis testing and large-deviations analysis, but unstable when the measures have very different tails.

## 6.5 The (Absence of) Coupling Underneath KL Divergence

Recall that the Wasserstein distance is defined as an infimum over the set $\Pi(\mu, \nu)$ of all couplings of $\mu$ and $\nu$.

Does $D_{\mathrm{KL}}(f \,\|\, g)$ come from a similar optimisation over $\Pi(\mu, \nu)$? In general, **no**. The KL divergence does not use a coupling of $f$ and $g$ at all. Its definition (1) requires only the two densities evaluated at the *same point* $x \in X$:

$$D_{\mathrm{KL}}(f \,\|\, g) = \int_X f(x) \log \frac{f(x)}{g(x)} \, d\lambda(x) = \mathbb{E}_\mu \left[ \log \frac{f(X)}{g(X)} \right].$$

There is only one random variable here, $X \sim \mu = f\lambda$; no second variable $Y \sim \nu$ appears. The comparison between $f$ and $g$ is *pointwise*: both densities are evaluated at the same location $x$ using the common reference measure $\lambda$, and their ratio $f(x)/g(x)$ is the log-likelihood ratio.

In informal terms: KL divergence compares the two mountains $f$ and $g$ by looking at their heights $f(x)$ and $g(x)$ at each location $x$. Wasserstein compares them by asking how much effort it takes to rearrange the earth in one mountain to create the other. KL cares about altitudes; Wasserstein cares about lateral displacement.

One way to see why no coupling naturally underlies KL: a coupling $\pi \in \Pi(\mu, \nu)$ requires the two marginals to be the *respective* $x$- and $y$-marginals. In KL, there is only one variable: both $f$ and $g$ are functions of the same variable $x$. The "natural" candidate for an underlying coupling—the *diagonal coupling* $\pi_\Delta(x, y) = f(x)\delta(y - x)$—has marginals $(\mu, \mu)$, not $(\mu, \nu)$, and so is not in $\Pi(\mu, \nu)$ unless $\mu = \nu$.

One can formally compute the KL divergence of the diagonal coupling from the independent coupling:

$$\begin{aligned} D_{\mathrm{KL}}(\pi_\Delta \,\|\, \mu \otimes \nu) &= \int_{X \times X} \pi_\Delta(x, y) \log \frac{\pi_\Delta(x, y)}{f(x) g(y)} \, d\lambda(x) \, d\lambda(y) \\ &= \int_X f(x) \log \frac{f(x)}{f(x) g(x)} \, d\lambda(x) = -\int_X f(x) \log g(x) \, d\lambda(x) = H_{\mathrm{cross}}(f, g), \end{aligned} \tag{23}$$

which is the *cross-entropy*, not $D_{\mathrm{KL}}(f \,\|\, g)$. The difference between cross-entropy and KL is precisely the entropy: $H_{\mathrm{cross}}(f, g) = H(f) + D_{\mathrm{KL}}(f \,\|\, g)$. So the diagonal coupling contributes the entropy term $H(f)$ as overhead, along with $D_{\mathrm{KL}}(f \,\|\, g)$. This shows

that even if one forces a coupling interpretation, there is an irreducible "entropic cost" $H(f)$ that has nothing to do with the discrepancy between $f$ and $g$.

**Summary.** The coupling perspective is fundamental to Wasserstein: $W_p = \inf_\pi[\cdots]$ over $\Pi(\mu,\nu)$. KL divergence has no such coupling: it is a functional of the ratio $f/g$ at the same point, using the common reference measure $\lambda$. The two objects are measuring fundamentally different things, and the question of "which coupling underlies KL" does not have a natural answer in the optimal transport sense.

## 6.6 Entropic Regularisation: KL as a Penalty on Couplings

Although KL divergence does not itself arise from optimising over $\Pi(\mu,\nu)$, it plays a central role as a *regulariser* when added to the Kantorovich objective. The *entropy-regularised optimal transport* problem, popularised by Cuturi [2013], is:

$$\boxed{\mathrm{OT}_\varepsilon(\mu,\nu) = \inf_{\pi\in\Pi(\mu,\nu)} \left[\int_{X\times X} c(x,y)\,d\pi(x,y) + \varepsilon\, D_{\mathrm{KL}}(\pi \,\|\, \mu\otimes\nu)\right], \quad \varepsilon > 0.} \tag{24}$$

Here $D_{\mathrm{KL}}(\pi \,\|\, \mu\otimes\nu)$ is the KL divergence of the joint $\pi$ from the *independent* coupling $\mu\otimes\nu$:

$$D_{\mathrm{KL}}(\pi \,\|\, \mu\otimes\nu) = \int_{X\times X} \pi(x,y)\log\frac{\pi(x,y)}{f(x)g(y)}\,d\lambda(x)\,d\lambda(y) = I_\pi(X;Y), \tag{25}$$

the *mutual information* between $X$ and $Y$ under the coupling $\pi$. The penalty $\varepsilon\, I_\pi(X;Y)$ discourages couplings that create strong statistical dependence between $X$ and $Y$; the independent coupling $\mu\otimes\nu$ achieves $I = 0$ and incurs no penalty.

The solution to (24) is a coupling $\pi^*_\varepsilon \in \Pi(\mu,\nu)$ with density

$$\pi^*_\varepsilon(x,y) \propto \exp\left(-\frac{c(x,y)}{\varepsilon}\right) f(x)g(y), \tag{26}$$

up to normalisation functions $u(x)$ and $v(y)$ (Sinkhorn scaling). The Sinkhorn algorithm [Cuturi, 2013] computes $u$ and $v$ by alternating projections, converging geometrically.

**Two limits of the regularisation parameter.**

1. **As** $\varepsilon \to 0$: $\pi_\varepsilon^*$ concentrates on the optimal transport plan of the unregularised problem (14), and $\mathrm{OT}_\varepsilon \to C(\mu, \nu; c)$ (the unregularised OT cost, i.e., $W_p^p$ when $c = d^p$). The penalty term vanishes, and the entropy of $\pi_\varepsilon^*$ tends to the (possibly small) entropy of the optimal transport plan.

2. **As** $\varepsilon \to \infty$: the transport-cost term becomes negligible relative to the KL penalty, and $\pi_\varepsilon^* \to \mu \otimes \nu$ (the independent coupling). Intuitively, when the penalty for statistical dependence is very large, the only affordable coupling is the independent one, regardless of how much transport cost it saves. The problem is dominated by entropy maximisation subject to the marginal constraints, and the solution is the independent coupling.

**Sinkhorn divergence.** The raw entropy-regularised cost $\mathrm{OT}_\varepsilon(\mu, \nu)$ is neither symmetric nor zero when $\mu = \nu$. The *Sinkhorn divergence* [Genevay et al., 2018] corrects this:

$$S_\varepsilon(\mu, \nu) = \mathrm{OT}_\varepsilon(\mu, \nu) - \tfrac{1}{2}\mathrm{OT}_\varepsilon(\mu, \mu) - \tfrac{1}{2}\mathrm{OT}_\varepsilon(\nu, \nu). \tag{27}$$

$S_\varepsilon$ is symmetric, non-negative, zero iff $\mu = \nu$, and interpolates between $W_p$ (as $\varepsilon \to 0$) and the squared maximum mean discrepancy (MMD) (as $\varepsilon \to \infty$). It enjoys the computational efficiency of entropy-regularised OT while retaining the metric flavour of Wasserstein.

## 6.7 The Schrödinger Bridge Problem

A related but distinct connection between couplings and KL divergence is the *Schrödinger bridge* problem, introduced by Schrödinger [1931] in the context of statistical mechanics.

Let $R \in \mathcal{P}(X \times X)$ be a *reference* joint distribution (not necessarily in $\Pi(\mu, \nu)$) that encodes some prior knowledge about how particles might pair up—for example, the joint distribution obtained from a Brownian bridge between two time points. The Schrödinger bridge problem asks:

$$\pi^* = \arg\min_{\pi \in \Pi(\mu,\nu)} D_{\mathrm{KL}}(\pi \,\|\, R)\,, \tag{28}$$

i.e., find the coupling that is closest to $R$ in KL divergence while having the prescribed marginals $\mu$ and $\nu$. This is a projection of $R$ onto the marginal-constrained set $\Pi(\mu, \nu)$ in the KL sense.

The Schrödinger bridge is notable for several reasons:

1. **KL on joints.** Unlike $W_p$, which minimises a transport cost $\int c\, d\pi$, the Schrödinger bridge minimises KL divergence between joint distributions. KL now appears as the objective, not a regulariser.

2. **Connection to entropic OT.** When $R(dx, dy) = e^{-c(x,y)/\varepsilon}\mu(dx)\nu(dy)/Z$ (i.e., when $R$ is the law of a pair with density proportional to the transport kernel), the Schrödinger bridge $\pi^*$ coincides with the entropy-regularised OT solution $\pi^*_\varepsilon$ in (26).

3. **Convergence to Monge.** When $R$ is the law of $(X_0, X_1)$ under a Brownian motion with diffusion coefficient $\sigma > 0$, as $\sigma \to 0$ the Schrödinger bridge converges to the Monge optimal transport plan.

A comprehensive survey of the Schrödinger bridge and its connections to optimal transport is given by Léonard [2014].

## 6.8 Inequalities Relating KL and Wasserstein

Even though KL divergence and Wasserstein distance measure different things, they can be bounded in terms of each other through intermediate quantities, principally the total variation distance.

**Pinsker's inequality.** The *total variation distance* between $\mu$ and $\nu$ is:

$$\|\mu - \nu\|_{\mathrm{TV}} = \sup_{A \in \mathcal{B}(X)} |\mu(A) - \nu(A)| = \frac{1}{2}\int_X |f(x) - g(x)|\, d\lambda(x). \tag{29}$$

Pinsker's inequality (see Cover and Thomas 2006) gives:

$$\boxed{\|\mu - \nu\|^2_{\mathrm{TV}} \leq \tfrac{1}{2}\, D_{\mathrm{KL}}(f \,\|\, g)\,.} \tag{30}$$

The reverse does not hold without additional conditions: KL can be large even when total variation is close to 1 (for distributions that barely overlap).

On a compact space with diameter $D = \sup_{x,y \in X} d(x, y)$, the Kantorovich-Rubinstein

theorem and the bound $\int h(f-g) \le \|h\|_\infty \|f-g\|_{L^1}$ give:

$$W_1(\mu,\nu) \le D\ \|\mu-\nu\|_{\mathrm{TV}}\,. \tag{31}$$

Combining (30) and (31):

$$W_1(\mu,\nu) \le D\sqrt{\tfrac{1}{2}\,D_{\mathrm{KL}}(f\,\|\,g)}. \tag{32}$$

Thus on bounded spaces, Wasserstein-1 is controlled by the square root of KL. The chain of inequalities is:

$$W_1 \;\le\; D\ \|\mu-\nu\|_{\mathrm{TV}} \;\le\; D\sqrt{\tfrac{1}{2}D_{\mathrm{KL}}(f\,\|\,g)}.$$

**Talagrand's transportation inequality.** The inequality (32) goes in the direction KL $\Rightarrow W_1$. For $W_2$, a related result holds under a structural condition on one of the measures.

A measure $\nu$ is said to satisfy a $T_2$ *transportation inequality* with constant $C > 0$ if for all $\mu \in \mathcal{P}(X)$:

$$W_2(\mu,\nu)^2 \le C\,D_{\mathrm{KL}}(\mu\,\|\,\nu)\,. \tag{33}$$

Talagrand [1996] proved (33) for $\nu$ the standard Gaussian on $\mathbb{R}^d$ with $C = 2$. The key insight is that the Gaussian satisfies a logarithmic Sobolev inequality (LSI) with constant $\alpha = 1$, and LSI implies $T_2$ with $C = 2/\alpha$.

The precise connection between LSI and $T_2$ was established by Otto and Villani [2000] through the *HWI inequality*:

$$D_{\mathrm{KL}}(\mu\,\|\,\nu) \le W_2(\mu,\nu)\sqrt{I(\mu|\nu)} - \frac{\alpha}{2}W_2(\mu,\nu)^2, \tag{34}$$

where $I(\mu|\nu) = \int f|\nabla \log(f/g)|^2\,d\lambda$ is the *relative Fisher information*, and $\alpha$ is the log-Sobolev constant of $\nu$. The inequality (34) is called HWI because it relates $H = D_{\mathrm{KL}}(\mu\,\|\,\nu)$ (entropy), $W = W_2$ (Wasserstein), and $I = I(\mu|\nu)$ (Fisher information). Combining the HWI inequality with Young's inequality yields Talagrand's $T_2$ from (33).

Thus under a log-Sobolev condition on $\nu$:

$$W_2(\mu,\nu)^2 \le \frac{2}{\alpha}\,D_{\mathrm{KL}}(\mu\,\|\,\nu)\,.$$

In the other direction, without further conditions, no inequality of the form $D_{\mathrm{KL}}(\mu \,\|\, \nu) \lesssim W_2(\mu,\nu)^\alpha$ holds globally on a compact space ($D_{\mathrm{KL}}$ can blow up while $W_2$ remains bounded).

Table 3 summarises the inequalities in one place.

Table 3: Inequalities relating KL divergence, total variation, and Wasserstein distance. Here $D = \mathrm{diam}(X)$ and $\alpha$ is the log-Sobolev constant of $\nu$.

| Inequality | Content | Reference |
|---|---|---|
| Pinsker | $\|\mu-\nu\|_{\mathrm{TV}}^2 \le \frac{1}{2} D_{\mathrm{KL}}(f \,\|\, g)$ | Cover and Thomas 2006 |
| $W_1$–TV | $W_1(\mu,\nu) \le D \, \|\mu-\nu\|_{\mathrm{TV}}$ | Kantorovich-Rubinstein |
| $W_1$–KL | $W_1(\mu,\nu) \le D\sqrt{\frac{1}{2} D_{\mathrm{KL}}(f \,\|\, g)}$ | Pinsker + KR |
| Talagrand $T_2$ | $W_2(\mu,\nu)^2 \le \frac{2}{\alpha} D_{\mathrm{KL}}(\mu \,\|\, \nu)$ | Talagrand 1996 |
| HWI | $D_{\mathrm{KL}}(\mu \,\|\, \nu) \le W_2\sqrt{I(\mu\|\nu)} - \frac{\alpha}{2} W_2^2$ | Otto and Villani 2000 |

**Remark on the sharpness of Pinsker.** Pinsker's inequality is not tight in general. For Gaussian distributions $\mu = \mathcal{N}(m_1, \sigma^2)$ and $\nu = \mathcal{N}(m_2, \sigma^2)$ with $\delta = |m_1 - m_2|/(2\sigma)$ (half the normalised mean shift): $D_{\mathrm{KL}}(\mu \,\|\, \nu) = \delta^2/2$ and $\|\mu-\nu\|_{\mathrm{TV}} = 2\Phi(\delta) - 1 \approx 2\phi(0)\delta$ for small $\delta$. So for small perturbations, $\|f-g\|_{\mathrm{TV}} \approx \frac{2}{\sqrt{2\pi}}\sqrt{2\, D_{\mathrm{KL}}(\mu \,\|\, \nu)}$, while Pinsker gives $\|f-g\|_{\mathrm{TV}} \le \sqrt{D_{\mathrm{KL}}(\mu \,\|\, \nu)\,/2}$. The two are of the same order but differ by a constant.

# 7 Conclusions

We summarise the main findings:

1. **KL divergence is insensitive to the ground metric on $X$, but induces its own geometry on $\mathcal{P}(X)$.** As an $f$-divergence, KL compares the two densities at the *same point* $x$ via the ratio $f(x)/g(x)$, using the common reference measure $\lambda$ as a shared coordinate system. No ground-metric cost $d(x,y)$ enters. However, KL is far from geometry-free: its Hessian generates the Fisher information metric on any statistical manifold $\{p(\cdot;\theta)\}$, the foundational object of Amari's information geometry [Amari, 2016, Amari and Nagaoka, 2000]. This is an intrinsic geometry

on the space of distributions, distinct from the ground-metric-derived $W_2$ (Otto) geometry. No coupling of $\mu$ and $\nu$ is involved in KL.

2. **Wasserstein distance is defined by optimising over couplings.** Every element of $\Pi(\mu, \nu)$ is a feasible coupling; the Wasserstein distance picks the cheapest one. The optimal coupling encodes the geometry-respecting "best matching" of mass in $\mu$ to mass in $\nu$.

3. **No coupling naturally underlies KL.** The "diagonal coupling" $\pi_\Delta(x, y) = f(x)\delta(y - x)$ has marginals $(\mu, \mu)$ rather than $(\mu, \nu)$, and so lies in $\Pi(\mu, \mu)$ rather than $\Pi(\mu, \nu)$. Its KL divergence from the product measure equals the cross-entropy $H_{\text{cross}}(f, g) = H(f) + D_{\text{KL}}(f \,\|\, g)$, not $D_{\text{KL}}(f \,\|\, g)$ itself.

4. **KL emerges as a regulariser on couplings.** In entropy-regularised OT (24), KL divergence of the joint $\pi$ from the independent coupling $\mu \otimes \nu$ appears as a penalty. This penalises couplings that create statistical dependence between the two marginals, and its role is to smooth the discrete optimal transport plan into a continuous kernel.

5. **KL emerges as an objective on joints in the Schrödinger bridge.** The Schrödinger bridge (28) minimises KL divergence of a coupling $\pi$ from a reference joint $R$, subject to marginal constraints. This is the natural dual to OT: where OT fixes a transport cost and optimises freely over $\Pi$, the Schrödinger bridge fixes a reference joint and optimises according to an entropic (KL) criterion.

6. **Inequalities connect the two.** On compact spaces, $W_1 \leq D\sqrt{D_{\text{KL}}/2}$ (via Pinsker and Kantorovich-Rubinstein), and $W_2^2 \leq (2/\alpha)D_{\text{KL}}$ (Talagrand) when $\nu$ satisfies a log-Sobolev inequality. These bounds go in the direction KL $\Rightarrow$ Wasserstein; the reverse typically fails without additional conditions.

7. **Both are special cases of a richer hierarchy.** The $f$-divergence family (of which KL is a member) and the class of Integral Probability Metrics (of which $W_1$ is a member) do not overlap. Their interaction is mediated by total variation distance, which belongs to both families. The entropy-regularised OT and the Schrödinger bridge occupy an intermediate position, blending variational optimisation over couplings with entropic functionals.

# A KL Divergence and Mutual Information

Given a coupling $\pi \in \Pi(\mu, \nu)$ with joint density $\pi(x, y)$, the *mutual information* between $X$ and $Y$ under $\pi$ is:

$$I_\pi(X;Y) = D_{\mathrm{KL}}(\pi \,\|\, \mu \otimes \nu) = \int_{X \times X} \pi(x,y) \log \frac{\pi(x,y)}{f(x)g(y)} \, d\lambda(x) \, d\lambda(y). \tag{35}$$

This is the KL divergence of the joint from the independent coupling. It measures how much statistical dependence the coupling $\pi$ introduces beyond what the two marginals alone would suggest.

Key properties of $I_\pi(X;Y)$:

- $I_\pi(X;Y) \geq 0$, with equality iff $X \perp Y$ under $\pi$ (i.e., $\pi = \mu \otimes \nu$).
- Among all couplings $\pi \in \Pi(\mu, \nu)$, the *independent coupling* minimises $I_\pi(X;Y)$, achieving the value 0. It is the least statistically dependent coupling.
- The *optimal transport coupling* $\pi_W^*$ (for $W_p$) typically maximises statistical dependence for a given marginal pair: the Monge map $Y = T(X)$ achieves $I_\pi(X;Y) = H(Y)$ (if $T$ is bijective and $\mu$ absolutely continuous), since $Y$ is fully determined by $X$ and $H(Y|X) = 0$.

The contrast is striking: the OT coupling maximises dependence (all information about $X$ is encoded in $Y$), while the independent coupling minimises it. The entropic regulariser $\varepsilon I_\pi$ in (24) penalises the OT coupling for being "too deterministic" and drives the solution toward the independent coupling as $\varepsilon$ increases.

# B The Special Case of Gaussian Distributions

Let $\mu = \mathcal{N}(m_1, \Sigma_1)$ and $\nu = \mathcal{N}(m_2, \Sigma_2)$ be Gaussian distributions on $\mathbb{R}^d$. Both the KL divergence and the Wasserstein-2 distance have closed forms.

**KL divergence.**

$$D_{\mathrm{KL}}(\mu \,\|\, \nu) = \tfrac{1}{2}\Big[\mathrm{tr}(\Sigma_2^{-1}\Sigma_1) + (m_2 - m_1)^\top \Sigma_2^{-1}(m_2 - m_1) - d + \log \frac{\det \Sigma_2}{\det \Sigma_1}\Big]. \tag{36}$$

**Wasserstein-2 distance (Bures metric).** The $W_2^2$ distance between Gaussians, sometimes called the Bures(-Wasserstein) metric, is [Villani, 2009]:

$$W_2(\mu,\nu)^2 = \|m_1 - m_2\|^2 + \mathcal{B}(\Sigma_1,\Sigma_2)^2, \tag{37}$$

where the Bures distance between covariances is

$$\mathcal{B}(\Sigma_1,\Sigma_2)^2 = \operatorname{tr}\Sigma_1 + \operatorname{tr}\Sigma_2 - 2\operatorname{tr}\Big[\big(\Sigma_1^{1/2}\Sigma_2\Sigma_1^{1/2}\big)^{1/2}\Big]. \tag{38}$$

For diagonal $\Sigma_1 = \sigma_1^2 I$ and $\Sigma_2 = \sigma_2^2 I$: $\mathcal{B}^2 = d(\sigma_1 - \sigma_2)^2$ and $W_2(\mu,\nu)^2 = \|m_1 - m_2\|^2 + d(\sigma_1 - \sigma_2)^2$.

The Gaussian case illustrates the geometric sensitivity of $W_2$: it depends linearly on the distance between means ($\|m_1 - m_2\|$) and on the distance between standard deviations ($|\sigma_1 - \sigma_2|$) in the isotropic case. The KL divergence (36) is not symmetric and grows quadratically in $\Sigma_2^{-1/2}(m_1 - m_2)$, favouring the direction of low variance in $\nu$.

# References


Shun-ichi Amari. *Differential-Geometrical Methods in Statistics*, volume 28 of *Lecture Notes in Statistics*. Springer, Berlin, 1985.

Shun-ichi Amari. *Information Geometry and Its Applications*, volume 194 of *Applied Mathematical Sciences*. Springer, Tokyo, 2016.

Shun-ichi Amari and Hiroshi Nagaoka. *Methods of Information Geometry*, volume 191 of *Translations of Mathematical Monographs*. American Mathematical Society and Oxford University Press, Providence, RI, 2000.

Luigi Ambrosio, Nicola Gigli, and Giuseppe Savaré. *Gradient Flows: In Metric Spaces and in the Space of Probability Measures*. Lectures in Mathematics ETH Zürich. Birkhäuser, Basel, 2nd edition, 2008.

Yann Brenier. Polar factorization and monotone rearrangement of vector-valued functions. *Communications on Pure and Applied Mathematics*, 44(4):375–417, 1991. doi: 10.1002/cpa.3160440402.

Thomas M. Cover and Joy A. Thomas. *Elements of Information Theory*. Wiley, Hoboken, NJ, 2nd edition, 2006.

Imre Csiszár. $I$-divergence geometry of probability distributions and minimization problems. *Annals of Probability*, 3(1):146–158, 1975. doi: 10.1214/aop/1176996454.

Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, pages 2292–2300, 2013.

Monroe D. Donsker and Srinivasa R. S. Varadhan. Asymptotic evaluation of certain Markov process expectations for large time, I. *Communications on Pure and Applied Mathematics*, 28(1):1–47, 1975. doi: 10.1002/cpa.3160280102.

Aude Genevay, Gabriel Peyré, and Marco Cuturi. Learning generative models with Sinkhorn divergences. In *Proceedings of the 21st International Conference on Artificial Intelligence and Statistics (AISTATS)*, volume 84, pages 1608–1617, 2018.

Richard Jordan, David Kinderlehrer, and Felix Otto. The variational formulation of the Fokker–Planck equation. *SIAM Journal on Mathematical Analysis*, 29(1):1–17, 1998. doi: 10.1137/S0036141096303359.

Leonid V. Kantorovich. On the translocation of masses. *Doklady Akademii Nauk SSSR*, 37:199–201, 1942.

Christian Léonard. A survey of the Schrödinger problem and some of its connections with optimal transport. *Discrete and Continuous Dynamical Systems*, 34(4):1533–1574, 2014. doi: 10.3934/dcds.2014.34.1533.

Robert J. McCann. A convexity principle for interacting gases. *Advances in Mathematics*, 128(1):153–179, 1997. doi: 10.1006/aima.1997.1634.

Frank Nielsen. An elementary introduction to information geometry. *Entropy*, 22(10): 1100, 2020.

Frank Nielsen and Richard Nock. Entropies and cross-entropies of exponential families. *Proceedings of the IEEE International Conference on Image Processing*, pages 3621–3624, 2010.

Felix Otto and Cédric Villani. Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *Journal of Functional Analysis*, 173(2): 361–400, 2000. doi: 10.1006/jfan.1999.3557.

Gabriel Peyré and Marco Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019. doi: 10.1561/2200000073.

C. Radhakrishna Rao. Information and the accuracy attainable in the estimation of statistical parameters. *Bulletin of the Calcutta Mathematical Society*, 37:81–91, 1945.

Erwin Schrödinger. Über die Umkehrung der Naturgesetze. *Sitzungsberichte der Preussischen Akademie der Wissenschaften, Physikalisch-mathematische Klasse*, pages 144–153, 1931.

Michel Talagrand. Transportation cost for Gaussian and other product measures. *Geometric and Functional Analysis*, 6(3):587–600, 1996. doi: 10.1007/BF02249265.

Cédric Villani. *Optimal Transport: Old and New*, volume 338 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, 2009.